

Nurul Islam Noman

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TECHNICAL SKILLS

- **Hardware & Engineering:** Embedded Electronics, Microcontrollers (STM32, Espressif/IoT), Circuit & Multi-layer PCB Design (Altium Designer, KiCad), 3D Modeling & Simulation (SolidWorks), Rapid Prototyping (SLS, SLA, FDM).
- **Software & Firmware:** Embedded C/C++, Python, Java, HTML/CSS, JavaScript, MATLAB.
- **Core Competencies:** Signal Integrity Analysis, Electromechanics, Flight Control Algorithms, Aerodynamics, Machine Learning (Edge).
- **Languages:** English (Fluent), Japanese (Beginner).

EXPERIENCE

[Drishti Technologies](#), Banani, Dhaka, Bangladesh

Feb 2024 - Jan 2025

Embedded System Engineer

On-site

- **Full-Lifecycle Product Development:** Planned and executed the complete end-to-end development of a Smart Glass for the visually impaired, featuring camera-based environmental analysis and an integrated navigation system.
- **PCB Design & Hardware Engineering:** Directed the design of multi-layer PCBs using Altium Designer for both the Smart Glass and the "ABHASH" intelligent white cane, executing schematic capture, complex routing, and rigorous signal integrity analysis.
- **Firmware Engineering:** Developed and optimized low-level microcontroller firmware (STM32/ESP) to drive critical product features, including camera integrations for the Smart Glass and real-time audio/haptic navigation systems for ABHASH.
- **Mechanical Integration & Prototyping:** Engineered precise 3D models and conducted structural simulations in SOLIDWORKS to validate the custom wearable enclosure for the Smart Glass, systematically troubleshooting bottlenecks to maximize system efficiency.

PROJECTS

[WRO Future Engineers Autonomous Car](#) | *Computer Vision, Sensor Fusion, Autonomous Navigation*

Nov 2025

- Engineered a self-driving vehicle with Team Echo Drift for the World Robot Olympiad (WRO) International Final in Singapore, securing 29th place globally by implementing vision-based track navigation and dynamic obstacle avoidance.

[SN-21 Folding-Wing VTOL UAV](#) | SolidWorks, Aerodynamics, Flight Control

Mar 2024 – Present

- Directed the 3D modeling and aerodynamic simulation of a Transwing VTOL system in SolidWorks to optimize structural integrity and flight mechanics.

[Autonomous Trash Collector Robot](#) | Electromechanics, Automation

Jan 2023

- Designed an automated robotic system featuring a custom-engineered electromechanical gripper for efficient object retrieval and transport.

[High-Speed Line Follower Robot](#) | Microcontrollers, PID Control, Sensor Integration

Feb 2022

- Engineered a high-performance, autonomous line-following robot.

EDUCATION

Bachelor of Science in Electrical and Electronics Engineering

University of Scholars | Dhaka, Bangladesh

Expected September 2028

Higher Secondary Certificate (Science)

MA Hashem University College | Begumgang Noakhali

2023

Secondary School Certificate

Chilonia Union High School | Senbag Noakhali

2021

HONORS AND AWARDS

- **Global Rank 29th**, Future Engineers Category – World Robot Olympiad Singapore, 2025
- **Champion**, Line Follower Robot Competition – UIU Feb 2025
- **2nd Place**, Autonomous Trash Collector Robot Competition – BUET Jan 2023
- **National Bronze Medalist**, Robot Gathering Competition – IRO-BD Nov 2021